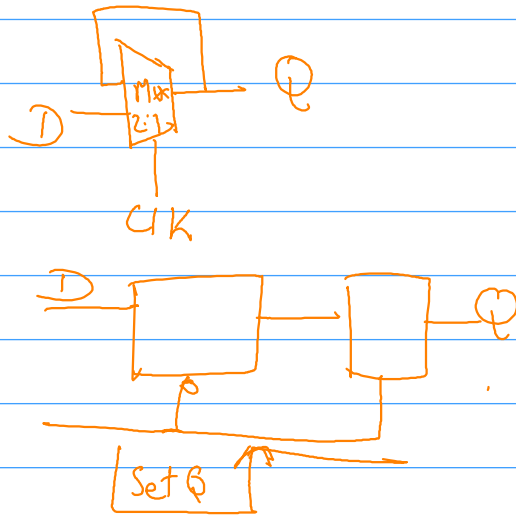


Assignment 2

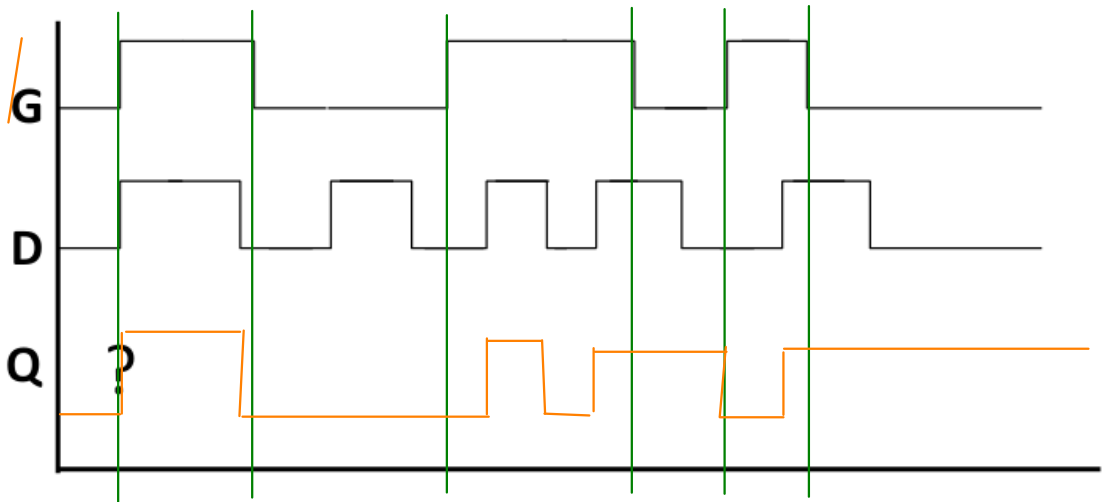
Q-1 D FF using 2:1 MUX.

⇒

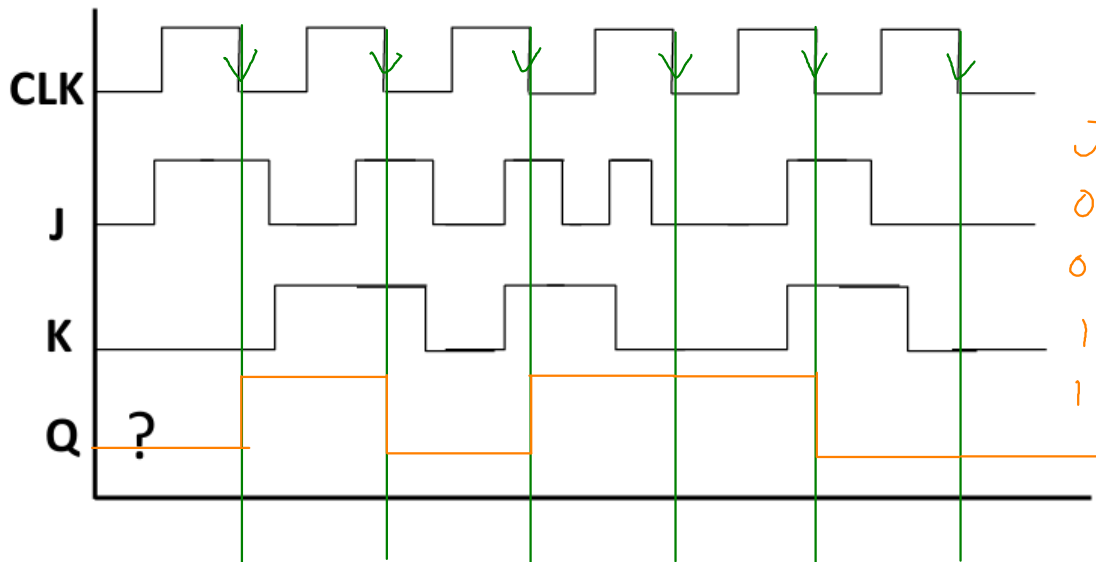


Q-2

2) Complete the below timing diagram for gated D Latch



3) Complete the below timing diagram for falling edge triggered JK FF.



J	K	Output
0	0	Init Q
0	1	Reset 0
1	0	Set 1
1	1	Toggle Q'

Q-4 Latch vs Flip flop

Controlled by enable

Edge triggered - Control by clk

Q-5 All flip flop conversion

Q	Q _{next}	SR	JK	D	T	
0	0	00	0,x	0	0	D: Q _{next} = D
0	1	10	1,x	1	1	T: Q _{next} = T ⊕ Q
1	0	01	x,1	0	1	JK: Q _{next} = JQ' + K'Q
1	1	11	x,0	1	0	SR: Q _{next} = S + RQ

Q-6 Convert JK Flip Flop to XY FF

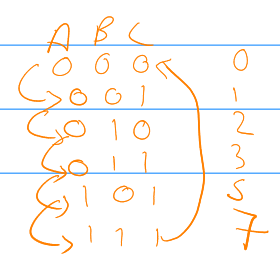
X	Y	Q
0	0	Q
0	1	Q
1	0	Q'
1	1	Q'

=>

TT of JK		
X	Y	Q
0	1	Q
0	0	Q
1	1	Q'
1	0	Q'

$J = X$
 $K = Y'$

Q-7 Design a counter which counts 0,1,2,3,5,7,0... using above XY FF and NOR gate



Max bit 3

My Name is Israr

Q	Q ⁺	X	Y
0	0	0	0
0	1	1	1
1	0	0	0
1	1	0	1

A B C	ABC ^{*(Next)}	X _A	Y _A	X _B	Y _B	X _C	Y _C
0 0 0	0 0 1	0	0	0	0	1	1
0 0 1	0 1 0	0	0	1	1	0	0
0 1 0	0 1 1	0	0	0	1	1	1
0 1 1	1 0 0	1	1	0	0	0	0
1 0 0	0 0 0	0	0	0	0	0	0
1 0 1	1 1 1	0	1	1	1	0	1
1 1 0	0 0 0	0	0	0	0	0	0
1 1 1	0 0 0	0	0	0	0	0	0

Unused
reset
counter

Solve X_A

A \ BC	00	01	11	10
0	0	0	1	0
1	0	0	0	0

$X_A = BC$

Solve Y_A

A \ BC	00	01	11	10
0	0	0	1	0
1	0	1	0	0

$Y_A = \bar{A} + \bar{B}C \approx A + \bar{B}\bar{C}$

Solve X_B

A \ BC	00	01	11	10
0	0	1	0	0
1	0	1	0	0

$X_B = C$

Solve Y_B

A \ BC	00	01	11	10
0	0	1	0	1
1	0	1	0	1

$Y_B = \bar{B} + \bar{C}$

Solve X_C

A \ BC	00	01	11	10
0	0	0	0	0
1	0	0	0	0

$X_C = \bar{A}$

Solve Y_C

A \ BC	00	01	11	10
0	1	0	0	1
1	0	1	0	0

$Y_C = \bar{C} + A\bar{B} \approx C + \bar{A}\bar{B}$

